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Antenna structure

Abstract

An antenna structure includes a metal mechanism element, a ground element, a feeding radiation element, and a dielectric substrate. The metal mechanism element has a slot. The ground element is coupled to the metal mechanism element. The feeding radiation element has a feeding point. The feeding radiation element is coupled to the ground element. The dielectric substrate has a first surface and a second surface which are opposite to each other. The feeding radiation element is disposed on the first surface of the dielectric substrate. The second surface of the dielectric substrate is adjacent to the metal mechanism element. The slot of the metal mechanism element is excited to generate a first frequency band and a second frequency band. The feeding radiation element is excited to generate a third frequency band. The ground element further includes a first protruding portion and a second protruding portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a Divisional Application of U.S. patent application Ser. No. 17/370,181 filed on Jul. 8, 2021 (now U.S. Pat. No. 11,831,086), which claims priority to Taiwan Patent Application No. 109137217 filed on Oct. 27, 2020, the entirety of which are incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The disclosure generally relates to an antenna structure, and more particularly, it relates to a multiband antenna structure.

Description of the Related Art

(2) With the advancements being made in mobile communication technology, mobile devices such as portable computers, mobile phones, multimedia players, and other hybrid functional portable electronic devices have become more common. To satisfy user demand, mobile devices can usually perform wireless communication functions. Some devices cover a large wireless communication area; these include mobile phones using 2G, 3G, and LTE (Long Term Evolution) systems and using frequency bands of 700 MHz, 850 MHz, 900 MHz, 1800 MHz, 1900 MHz, 2100 MHz, 2300 MHz, and 2500 MHz. Some devices cover a small wireless communication area; these include mobile phones using Wi-Fi and Bluetooth systems and using frequency bands of 2.4 GHz, 5.2 GHz, and 5.8 GHz.

(3) Antennas are indispensable elements for wireless communication. If an antenna used for signal reception and transmission has insufficient bandwidth, the communication quality of the mobile device will suffer. Accordingly, it has become a critical challenge for antenna designers to design a small wideband antenna element.

BRIEF SUMMARY OF THE INVENTION

(4) In an exemplary embodiment, the disclosure is directed to an antenna structure that includes a metal mechanism element, a ground element, a feeding radiation element, and a dielectric substrate. The metal mechanism element has a slot. The slot has a first closed end and a second closed end. The ground element is coupled to the metal mechanism element. The feeding radiation element has a feeding point. The feeding radiation element is coupled to the ground element. The dielectric substrate has a first surface and a second surface which are opposite to each other. The feeding radiation element is disposed on the first surface of the dielectric substrate. The second surface of the dielectric substrate is adjacent to the metal mechanism element. The slot of the metal

mechanism element is excited to generate a first frequency band and a second frequency band. The feeding radiation element is excited to generate a third frequency band. The ground element further includes a first protruding portion and a second protruding portion.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The invention can be more fully understood by reading the subsequent detailed description and examples with references made to the accompanying drawings, wherein:
- (2) FIG. 1 is a top view of an antenna structure according to an embodiment of the invention;
- (3) FIG. 2 is a sectional view of an antenna structure according to an embodiment of the invention;
- (4) FIG. 3 is a diagram of VSWR (Voltage Standing Wave Ratio) of an antenna structure according to an embodiment of the invention;
- (5) FIG. 4 is a diagram of radiation efficiency of an antenna structure according to an embodiment of the invention;
- (6) FIG. 5 is a top view of an antenna structure according to an embodiment of the invention; and
- (7) FIG. 6 is a sectional view of an antenna structure according to an embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

- (8) In order to illustrate the purposes, features and advantages of the invention, the embodiments and figures of the invention are shown in detail as follows.
- (9) Certain terms are used throughout the description and following claims to refer to particular components. As one skilled in the art will appreciate, manufacturers may refer to a component by different names. This document does not intend to distinguish between components that differ in name but not function. In the following description and in the claims, the terms “include” and “comprise” are used in an open-ended fashion, and thus should be interpreted to mean “include, but not limited to . . .”. The term “substantially” means the value is within an acceptable error range. One skilled in the art can solve the technical problem within a predetermined error range and achieve the proposed technical performance. Also, the term “couple” is intended to mean either an indirect or direct electrical connection. Accordingly, if one device is coupled to another device, that connection may be through a direct electrical connection, or through an indirect electrical connection via other devices and connections.
- (10) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
- (11) Furthermore, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(12) FIG. 1 is a top view of an antenna structure **100** according to an embodiment of the invention. FIG. 2 is a sectional view of the antenna structure **100** according to an embodiment of the invention (along a sectional line LC1 of FIG. 1). Please refer to FIG. 1 and FIG. 2 together. The antenna structure **100** may be applied to a mobile device, such as a smart phone, a tablet computer, or a notebook computer. In the embodiment of FIG. 1 and FIG. 2, the antenna structure **100** includes a metal mechanism element **110**, a ground element **130**, a feeding radiation element **140**, and a dielectric substrate **180**. The ground element **130** and the feeding radiation element **140** may both be made of metal materials, such as copper, silver, aluminum, iron, or their alloys.

(13) The metal mechanism element **110** may be a metal housing of a mobile device. In some embodiments, the metal mechanism element **110** is a metal upper cover of a notebook computer, or a metal back cover of a tablet computer, but it is not limited thereto. For example, if the mobile device is a notebook computer, the metal mechanism element **110** may be the so-called “A-component” in the field of notebook computers. The metal mechanism element **110** has a slot **120**. The slot **120** of the metal mechanism element **110** may substantially have a straight-line shape. Specifically, the slot **120** has a first closed end **121** and a second closed end **122** which are away from each other. The antenna structure **100** may also include a nonconductive material which fills the slot **120** of the metal mechanism element **110**, so as to achieve the waterproof or dustproof function.

(14) The dielectric substrate **180** may be an FR4 (Flame Retardant 4) substrate, a PCB (Printed Circuit Board), or an FPC (Flexible Printed Circuit). The dielectric substrate **180** has a vertical projection on the metal mechanism element **110**, and the vertical projection can cover the whole slot **120** of the metal mechanism element **110**. The dielectric substrate **180** has a first surface E1 and a second surface E2 which are opposite to each other. The feeding radiation element **140** is disposed on the first surface E1 of the dielectric substrate **180**. The second surface E2 of the dielectric substrate **180** is adjacent to the slot **120** of the metal mechanism element **110**. It should be noted that the term “adjacent” or “close” over the disclosure means that the distance (spacing) between two corresponding elements is smaller than a predetermined distance (e.g., 5 mm or shorter), or means that the two corresponding elements directly touch each other (i.e., the aforementioned distance/spacing therebetween is reduced to 0). In some embodiments, the second surface E2 of the dielectric substrate **180** is directly attached to the metal mechanism element **110**.

(15) The ground element **130** may be a ground copper foil, which may be couple to the metal mechanism element **110**. In some embodiments, the ground element **130** extends from the metal mechanism element **110** onto the first surface E1 of the dielectric substrate **180**.

(16) The feeding radiation element **140** may substantially have a T-shape, and its vertical projection can at least partially overlap the slot **120** of the metal mechanism element **110**. The feeding radiation element **140** includes a first branch **150**, a second branch **160**, and a third branch **170**. The first branch **150** and the third branch **170** are both coupled through the second branch **160** to the ground element **130**. The first branch **150** may substantially have a wide straight-line shape, which may be substantially parallel to the ground element **130**. Specifically, the first branch **150** has a first end **151** and a second end **152**. A feeding point FP1 is positioned at the first end **151** of the first branch **150**. The feeding point FP1 may be coupled to a signal source (not shown). For example, the aforementioned signal source may be an RF (Radio Frequency) module for exciting the antenna structure **100**.

(17) The second branch **160** may substantially have a parallelogram or a rectangular shape. Specifically, the second branch **160** has a first end **161** and a second end **162**. The first end **161** of the second branch **160** is coupled to the ground element **130**. The second end **162** of the second branch **160** is coupled to the second end **152** of the first branch **150**. The third branch **170** may substantially have a narrow straight-line shape (narrower than the first branch **150**), which may be substantially parallel to the ground element **130**. Specifically, the third branch **170** has a first end **171** and a second end **172**. The first end **171** of the third branch **170** is coupled to the second end

162 of the second branch **160**. The second end **172** of the third branch **170** is an open end. The second end **172** of the third branch **170** and the first end **151** of the first branch **150** may substantially extend in opposite directions and away from each other. In some embodiments, the angle θ between the second branch **160** and the third branch **170** is from 90 to 180 degrees (e.g., about 120 degrees). However, the invention is not limited thereto. In another embodiment, the aforementioned angle θ is changed to be exactly 90 degrees, such that the second branch **160** and the third branch **170** are perpendicular to each other. It should be noted that the first branch **150** and the third branch **170** have vertical projections on the metal mechanism element **110**, and the whole vertical projections are inside the slot **120**.

(18) FIG. 3 is a diagram of VSWR (Voltage Standing Wave Ratio) of the antenna structure **100** according to an embodiment of the invention. The horizontal axis represents operation frequency (MHz), and the vertical axis represents the VSWR. According to the measurement of FIG. 3, the antenna structure **100** covers a first frequency band FB1, a second frequency band FB2, and a third frequency band FB3. For example, the first frequency band FB1 may be from 2400 MHz to 2500 MHz, the second frequency band FB2 may be from 5150 MHz to 5850 MHz, and the third frequency band FB3 may be from 6000 MHz to 7125 MHz. Thus, the antenna structure **100** can support at least the wideband operations of the conventional WLAN (Wireless Local Area Network) 2.4 GHz/5 GHz and the next generation Wi-Fi 6E.

(19) With respect to the antenna theory, the slot **120** of the metal mechanism element **110** can be excited by the feeding radiation element **140** using a coupling mechanism, so as to generate the first frequency band FB1 and the second frequency band FB2. The first branch **150** and the second branch **160** of the feeding radiation element **140** are excited to generate the third frequency band FB3. On the other hand, the third branch **170** of the feeding radiation element **140** is configured to fine-tune the impedance matching of the first frequency band FB1, the second frequency band FB2, and the third frequency band FB3. According to practical measurements, if the whole vertical projections of the first branch **150** and the third branch **170** of the feeding radiation element **140** are designed inside the slot **120**, the operation bandwidth of the antenna structure **100** is effectively increased.

(20) FIG. 4 is a diagram of radiation efficiency of the antenna structure **100** according to an embodiment of the invention. The horizontal axis represents operation frequency (MHz), and the vertical axis represents the radiation efficiency (%). According to the measurement of FIG. 4, the radiation efficiency of the antenna structure **100** is higher than 25% within the first frequency band FB1, the second frequency band FB2, and the third frequency band FB3. It can meet the requirements of practical application of general mobile communication devices.

(21) In some embodiments, the sizes of the elements of the antenna structure **100** are as follows. The distance D1 between the feeding point FP1 and the first closed end **121** of the slot **120** may be from 0.25 to 0.5 wavelength ($\lambda/4 \sim \lambda/2$) of the first frequency band FB1 of the antenna structure **100**. The distance D2 between the feeding point FP1 and the second closed end **122** of the slot **120** may be from 0.25 to 0.5 wavelength ($\lambda/4 \sim \lambda/2$) of the second frequency band FB2 of the antenna structure **100**. The width W1 of the slot **120** may be from 2 mm to 3 mm. The total length L1 of the first branch **150** and the second branch **160** may from 0.25 to 0.5 wavelength ($\lambda/4 \sim \lambda/2$) of the third frequency band FB3 of the antenna structure **100**. The length L2 of the third branch **170** may be from 4 mm to 5 mm. The width W2 of the first branch **150** may be at least twice the width W3 of the third branch **170**. The distance D3 between the first branch **150** and the ground element **130** may be from 1 mm to 1.5 mm. The distance D4 between the third branch **170** and the ground element **130** may be from 1.5 mm to 2 mm. The thickness H1 of the dielectric substrate **180** may be about 0.2 mm. The above ranges of element sizes are calculated and obtained according to many experiment results, and they help to optimize the operation bandwidth and impedance matching of the antenna structure **100**.

(22) FIG. 5 is a top view of an antenna structure **500** according to an embodiment of the invention.

FIG. 6 is a sectional view of the antenna structure **500** according to an embodiment of the invention (along a sectional line LC2 of FIG. 5). Please refer to FIG. 5 and FIG. 6 together. In the embodiment of FIG. 5 and FIG. 6, the antenna structure **500** includes a metal mechanism element **510**, a ground element **530**, a feeding radiation element **540**, and a dielectric substrate **580**. The ground element **530** and the feeding radiation element **540** may both be made of metal materials. (23) The metal mechanism element **510** may be a metal housing of a mobile device. In some embodiments, the metal mechanism element **510** is a metal upper cover of a notebook computer, or a metal back cover of a tablet computer, but it is not limited thereto. The metal mechanism element **510** has a slot **520**. The slot **520** of the metal mechanism element **510** may substantially have a straight-line shape. Specifically, the slot **520** has a first closed end **521** and a second closed end **522** which are away from each other. The antenna structure **500** may also include a nonconductive material which fills the slot **520** of the metal mechanism element **510**, so as to achieve the waterproof or dustproof function.

(24) The dielectric substrate **580** may be an FR4 substrate, a PCB, or an FPC. The dielectric substrate **580** has a vertical projection on the metal mechanism element **510**, and the vertical projection can cover the whole slot **520** of the metal mechanism element **510**. The dielectric substrate **580** has a first surface E3 and a second surface E4 which are opposite to each other. The feeding radiation element **540** is disposed on the first surface E3 of the dielectric substrate **580**. The second surface E4 of the dielectric substrate **580** is adjacent to the slot **520** of the metal mechanism element **510**. In some embodiments, the second surface E4 of the dielectric substrate **580** is directly attached to the metal mechanism element **510**.

(25) The ground element **530** may be a ground copper foil, which may be couple to the metal mechanism element **510**. In some embodiments, the ground element **530** extends from the metal mechanism element **510** onto the first surface E3 of the dielectric substrate **580**. Specifically, the ground element **530** further includes a first protruding portion **534** and a second protruding portion **535**, which may be both disposed on the first surface E3 of the dielectric substrate **580**. For example, the first protruding portion **534** of the ground element **530** may substantially have a straight-line shape, and the second protruding portion **535** of the ground element **530** may substantially have an inverted T-shape.

(26) The feeding radiation element **540** may substantially have an M-shape, and its vertical projection can at least partially overlap the slot **520** of the metal mechanism element **510**. Specifically, the feeding radiation element **540** has a first end **541** and a second end **542**. A feeding point FP2 is positioned at the first end **541** of the feeding radiation element **540**. The feeding point FP2 may be coupled to a signal source. For example, the aforementioned signal source may be an RF module for exciting the antenna structure **500**. In some embodiments, the feeding radiation element **540** includes a first rectangular widening portion **544**, a second rectangular widening portion **545**, and a third rectangular widening portion **546**. The first rectangular widening portion **544** is adjacent to the first protruding portion **534** of the ground element **530**, such as a coupling gap GC1 is formed therebetween. In addition, the third rectangular widening portion **546** is positioned at the second end **542** of the feeding radiation element **540**. The second rectangular widening portion **545** is positioned between the first rectangular widening portion **544** and the third rectangular widening portion **546**. It should be noted that the first rectangular widening portion **544**, the second rectangular widening portion **545**, and the third rectangular widening portion **546** have vertical projections on the metal mechanism element **510**, and the whole vertical projections are inside the slot **520**. Furthermore, the feeding point FP2 may be positioned between the first protruding portion **534** and the second protruding portion **535** of the ground element **530**.

(27) In some embodiments, the antenna structure **500** further includes a circuit component **550**. The circuit component **550** may be a fixed capacitor, a fixed inductor, or a fixed resistor. Alternatively, the circuit component **550** may be a variable capacitor, a variable inductor, or a variable resistor, whose impedance value is adjustable according to a control voltage of a processor. The third

rectangular widening portion **546** of the feeding radiation element **540** may be further coupled through the circuit component **550** to the second protruding portion **535** of the ground element **530**.

(28) According to the practical measurement, the antenna structure **500** covers a first frequency band, a second frequency band, and a third frequency band. For example, the aforementioned first frequency band may be from 2400 MHz to 2500 MHz, the aforementioned second frequency band may be from 5150 MHz to 5850 MHz, and the aforementioned third frequency band may be from 6000 MHz to 7125 MHz. Thus, the antenna structure **500** can support at least the wideband operations of the conventional WLAN 2.4 GHz/5 GHz and the next generation Wi-Fi **6E**.

(29) With respect to the antenna theory, the slot **520** of the metal mechanism element **510** can be excited by the feeding radiation element **540** using a coupling mechanism, so as to generate the aforementioned first and second frequency bands. The feeding radiation element **540** is excited to generate the aforementioned third frequency band. According to practical measurements, the incorporation of the first rectangular widening portion **544**, the second rectangular widening portion **545**, and the third rectangular widening portion **546** can fine-tune the impedance matching of the aforementioned first, second and third frequency bands. Moreover, the incorporation of the circuit component **550** can help to reduce the total size of the antenna structure **500**.

(30) In some embodiments, the sizes of the elements of the antenna structure **500** are as follows. The distance **D5** between the feeding point **FP2** and the first closed end **521** of the slot **520** may be from 0.25 to 0.5 wavelength ($\lambda/4 \sim \lambda/2$) of the first frequency band of the antenna structure **500**. The distance **D6** between the feeding point **FP2** and the second closed end **522** of the slot **520** may be from 0.25 to 0.5 wavelength ($\lambda/4 \sim \lambda/2$) of the second frequency band of the antenna structure **500**. The width **W4** of the slot **520** may be from 2 mm to 3 mm. The length **L3** of the feeding radiation element **540** may from 0.25 to 0.5 wavelength ($\lambda/4 \sim \lambda/2$) of the third frequency band of the antenna structure **500**. The width **W5** of the first rectangular widening portion **544** may be greater than the width **W6** of the second rectangular widening portion **545**. The width **W6** of the second rectangular widening portion **545** may be greater than the width **W7** of the third rectangular widening portion **546**. For example, the width **W5** may be from 5 mm to 7 mm, the width **W6** may be from 3 mm to 5 mm, and the width **W7** may be from 2 mm to 3 mm. The distance **D7** between the first rectangular widening portion **544** and the second rectangular widening portion **545** may be from 1 mm to 2 mm. The distance **D8** between the second rectangular widening portion **545** and the third rectangular widening portion **546** may be from 3 mm to 4 mm. The width of the coupling gap **GC1** may be smaller than 0.5 mm. The circuit component **550** may be a capacitor, whose capacitance may be from 0.1 pF to 2 pF, such as about 0.9 pF. The thickness **H2** of the dielectric substrate **580** may be about 0.2 mm. The above ranges of element sizes are calculated and obtained according to many experiment results, and they help to optimize the operation bandwidth and impedance matching of the antenna structure **500**.

(31) The invention proposes a novel antenna structure for integrating with a metal mechanism element of a mobile device. In comparison to the conventional design, the invention has at least the advantages of small size, wide bandwidth, low manufacturing cost, and beautiful device appearance, and therefore it is suitable for application in a variety of mobile communication devices.

(32) Note that the above element sizes, element shapes, element parameters, and frequency ranges are not limitations of the invention. An antenna designer can fine-tune these settings or values according to different requirements. It should be understood that the antenna structure of the invention is not limited to the configurations of FIGS. **1-6**. The invention may merely include any one or more features of any one or more embodiments of FIGS. **1-6**. In other words, not all of the features displayed in the figures should be implemented in the antenna structure of the invention.

(33) Use of ordinal terms such as “first”, “second”, “third”, etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as

labels to distinguish one claim element having a certain name from another element having the same name (but for use of the ordinal term) to distinguish the claim elements.

(34) While the invention has been described by way of example and in terms of the preferred embodiments, it should be understood that the invention is not limited to the disclosed embodiments. On the contrary, it is intended to cover various modifications and similar arrangements (as would be apparent to those skilled in the art). Therefore, the scope of the appended claims should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements.

Claims

1. An antenna structure, comprising: a metal mechanism element, having a slot, wherein the slot has a first closed end and a second closed end; a ground element, coupled to the metal mechanism element; a feeding radiation element, having a feeding point, wherein the feeding radiation element is coupled to the ground element; and a dielectric substrate, having a first surface and a second surface opposite to each other, wherein the feeding radiation element is disposed on the first surface, and the second surface is adjacent to the metal mechanism element; wherein the slot of the metal mechanism element is excited to generate a first frequency band and a second frequency band, and the feeding radiation element is excited to generate a third frequency band; wherein the ground element further comprises a first protruding portion and a second protruding portion; wherein the feeding radiation element comprises a first rectangular widening portion, a second rectangular widening portion, and a third rectangular widening portion, and the first rectangular widening portion is adjacent to the first protruding portion of the ground element; wherein the antenna structure further comprises: a circuit component, wherein the third rectangular widening portion is coupled through the circuit component to the second protruding portion of the ground element.
 2. The antenna structure as claimed in claim 1, wherein the first frequency band is from 2400 MHz to 2500 MHz, the second frequency band is from 5150 MHz to 5850 MHz, and the third frequency band is from 6000 MHz to 7125 MHz.
 3. The antenna structure as claimed in claim 1, wherein a distance between the feeding point and the first closed end of the slot is from 0.25 to 0.5 wavelength of the first frequency band.
 4. The antenna structure as claimed in claim 1, wherein a distance between the feeding point and the second closed end of the slot is from 0.25 to 0.5 wavelength of the second frequency band.
 5. The antenna structure as claimed in claim 1, wherein the feeding radiation element substantially has an M-shape.
 6. The antenna structure as claimed in claim 1, wherein a length of the feeding radiation element is from 0.25 to 0.5 wavelength of the third frequency band.
 7. The antenna structure as claimed in claim 1, wherein the first protruding portion of the ground element substantially has a straight-line shape.
 8. The antenna structure as claimed in claim 1, wherein the second protruding portion of the ground element substantially has an inverted T-shape.
 9. The antenna structure as claimed in claim 1, wherein the first rectangular widening portion, the second rectangular widening portion, and the third rectangular widening portion have vertical projections on the metal mechanism element, and the vertical projections are inside the slot.
 10. The antenna structure as claimed in claim 1, wherein the circuit component is a capacitor.
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